Every Algebraic Number is a Critical Value of an Integer Polynomial

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Abstract

For every nonconstant $f \in \mathbb{Z}[x]$ we construct $g \in \mathbb{Z}[x]$ and an irreducible $H \in \mathbb{Z}[x]$ with $H \mid g'$ and $H^2 \mid f \circ g$. Equivalently, every algebraic number is a critical value of an integer polynomial, so the set of critical values of $\mathbb{Z}[x]$ is exactly $\overline{\mathbb{Q}}$. The construction is explicit. The proof was assisted by AI, and has been formalized in Lean.

1 Introduction

A critical value of a nonconstant $g \in \mathbb{Z}[x]$ is a number α with $g(\beta) = \alpha$ and $g'(\beta) = 0$ for some β. Since $\overline{\mathbb{Q}}$ is closed under polynomial evaluation, every critical value is algebraic. This note proves the converse, and does so with an explicit construction.

Theorem 1. *Let $f \in \mathbb{Z}[x]$ be nonconstant. Then there exist $g, H \in \mathbb{Z}[x]$ with H irreducible in $\mathbb{Z}[x]$ and $1 \leq \deg H$, such that*

$$H \mid g' \quad \textit{and} \quad H^2 \mid f \circ g \qquad \textit{in } \mathbb{Z}[x].$$

The g produced satisfies $\deg g \geq 2$.

The corollary is the statement in the title.

Corollary 1. *The set of critical values is exactly $\overline{\mathbb{Q}}$.*

2 The construction

Let $h \in \mathbb{Z}[x]$ be a primitive irreducible factor of f, of degree n, with root θ; write $h(x) = \sum_{k=0}^{n} a_k x^{n-k}$, so that $a_n = h(0)$. If $h = x$, then $g = x^2$ and $H = x$ work. Otherwise $h(0) \neq 0$, and since $\gcd(h, h') = 1$ in $\mathbb{Q}[x]$, the elimination ideal $(h, h') \cap \mathbb{Z}$ is nonzero; let $u, v \in \mathbb{Z}[x]$ and $\rho \in \mathbb{Z}$, $\rho \neq 0$, be such that

$$u\,h + v\,h' = \rho,$$

and set

$$M = \rho\, a_n, \qquad H(x) = \frac{h(Mx)}{a_n} = \sum_{k=0}^{n} A_k x^{n-k}, \qquad W(x) = \frac{v(0)H(x) - v(Mx)}{\rho}.$$

The polynomial claimed by Theorem 1 is

$$g(x) = Mx + h(Mx)\,W(x), \tag{1}$$

together with this H.

Theorem (restatement of Theorem 1). *H is irreducible in $\mathbb{Z}[x]$, and $H \mid g'$ and $H^2 \mid f \circ g$ in $\mathbb{Z}[x]$.*

pf. From $A_k = a_k M^{n-k}/a_n$ we get $A_n = 1$ and $A_k = a_k \rho^{n-k} a_n^{n-k-1}$ for $k < n$, so $H \in \mathbb{Z}[x]$ with $H(0) = 1$, which gives $W(0) = 0$. And ρ divides M and every A_k with $k < n$, so it divides $v(0)H(x) - v(Mx)$ and $W \in \mathbb{Z}[x]$. Therefore g is an integral polynomial.

Let $\beta = \theta/M$, a root of H. Then $W(\beta) = -v(\theta)/\rho = -1/h'(\theta)$, so

$$g(\beta) = \theta, \qquad g'(\beta) = M(1 + h'(\theta)W(\beta)) = 0.$$

Hence β is a double root of $f \circ g$, and H is primitive irreducible, so $H \mid g'$ and $H^2 \mid f \circ g$ in $\mathbb{Z}[x]$ by Gauss. ∎

The computation uses $h(\theta) = 0$ and nothing else about θ, so it runs at every root of h at once: all n conjugates are critical values of the same g, at the n points θ_i/M. That is what makes H^2, and not merely $(x - \beta)^2$, divide $f \circ g$.

Any ρ will do, but $M = \rho\, a_n$ multiplies into every coefficient of H, W and g, so the smallest gives the smallest answer: the generator of $(h, h') \cap \mathbb{Z}$, the reduced resultant, of which $\mathrm{Res}(h, h')$ is a multiple. On $h = x^2 + 1$ the generator is 2 against a resultant of 4, and on Dedekind's cubic below 1006 against 2012.

3 Examples

Each line below is a closed identity, checkable by expansion.

Example 1 ($f = qx - p$, $p \neq 0$, $\gcd(p, q) = 1$). Here $h' = q$ and $p = qx - h$, so $(h, h') \cap \mathbb{Z} = \mathbb{Z}$ and $\rho = 1$. Choose $a, b \in \mathbb{Z}$ with $ap + bq = 1$ and take $u = -a$, $v = ax + b$; then $h(0) = -p$, $M = -p$, $H = qx + 1$, $W = x$, and

$$g = -pqx^2 - 2px, \qquad g' = -2p\,(qx + 1), \qquad f \circ g = -p\,(qx + 1)^2.$$

Example 2 ($f = x^2 + 1$). With $u = 2$, $v = -x$, $\rho = 2$, $h(0) = 1$, $M = 2$, $H = 4x^2 + 1$ and $W = x$,

$$g = 4x^3 + 3x, \qquad g' = 3(4x^2 + 1), \qquad f \circ g = (4x^2 + 1)^2(x^2 + 1).$$

Here $\deg g = 3 = n + 1$, the least degree any solution can have. The unit group $\mathcal{O}_K^\times = \{\pm 1, \pm i\}$ has rank 0, and it is $\mathbb{Z}[i/2] = \mathbb{Z}[i][1/2]$ that supplies the needed unit $1 + i$.

Example 3 ($f = x^3 - 2$). With $u = 3$, $v = -x$, $\rho = -6$, $h(0) = -2$, $M = 12$, $H = 1 - 864x^3$ and $W = -2x$,

$$g = -3456x^4 + 16x, \qquad g' = 16\,(1 - 864x^3).$$

Example 4 ($f = x^3 - x - 1$). The construction gives $M = 23$ and a sextic. A hand-tuned quintic also witnesses the *statement*, and is a useful independent check of it:

$$g = -24334x^5 + 39675x^4 - 26450x^3 + 9085x^2 - 1610x + 118, \qquad H = 23x^3 - 23x^2 + 8x - 1,$$

with $g' = -230\,(23x - 7)H$ and $H^2 \mid f \circ g$, the cofactor having degree 9.

Example 5 (Dedekind's cubic). $f = x^3 - x^2 - 2x - 8$, the standard example of a field whose ring of integers is not monogenic: 2 is a common index divisor [Ded78]. With $u = 21x - 125$, $v = -7x^2 + 44x - 3$, $\rho = 1006$, $h(0) = -8$, $M = -8048$,

$$H = 65158925824x^3 + 8096288x^2 - 2012x + 1, \qquad W = -194310912x^3 + 426544x^2 + 358x,$$

$$\begin{aligned} g = {}& 101288722414414331904x^6 - 209759614012620800x^5 \\ & - 217370176548864x^4 - 14767629312x^3 + 2350016x^2 - 10912x. \end{aligned}$$

4 Formalization

The development above has been formalized in Lean 4 [MU21] against Mathlib [The20], in about 1,200 lines. The main statement is

```
theorem main (f : Z[X]) (hf : 1 <= f.natDegree) :
    exists g H : Z[X], Irreducible H /\ H | derivative g /\ H^2 | f.comp g
```

with a strengthened form carrying $2 \leq \deg g$ and $1 \leq \deg H$, and with Corollary 1 proved from it. Every result depends only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

References

[Ded78] Richard Dedekind. "Über den Zusammenhang zwischen der Theorie der Ideale und der Theorie der höheren Kongruenzen". In: *Abhandlungen der Königlichen Gesellschaft der Wissenschaften zu Göttingen* (1878).

[The20] The mathlib Community. "The Lean Mathematical Library". In: *Certified Programs and Proofs (CPP 2020)*. ACM, 2020.

[MU21] Leonardo de Moura and Sebastian Ullrich. "The Lean 4 Theorem Prover and Programming Language". In: *Automated Deduction – CADE 28*. Lecture Notes in Computer Science. Springer, 2021.